User manual addons

This manual covers all currently available Spline Architect addons.

The currently available official addons are:

- Terrain Tools
- Object Cloning

No other official addons exist at this time.

Addons are separate from the main Spline Architect package and can be deleted from the Addons folder if not needed. Note that removing an addon will also remove all data created by that addon.

Installation

> ❗ You need to have the base package Spline Architect installed before installing any addon.

Go to Window → Package Manager → My Assets, install your Spline Architect addon, and you're done.

If you've moved the main Spline Architect folder, you will get some errors. This is because the addon assumes the SplineArchitect folder is located in the Assets folder. You can easily fix this by moving the addon from Assets/SplineArchitect/Addons/YourAddon to Assets/YourPath/SplineArchitect/Addons/YourAddon.

Updating

> ❗ If you get errors after updating an addon, remember to also check for updates to the base Spline Architect package.

To update, open Unity and go to Window → Package Manager → My Assets, then update the related package.

Terrain Tools

The terrain tools addon has 3 main features.

- Deform terrain.
- Paint terrain.
- Instantiate objects along a spline.

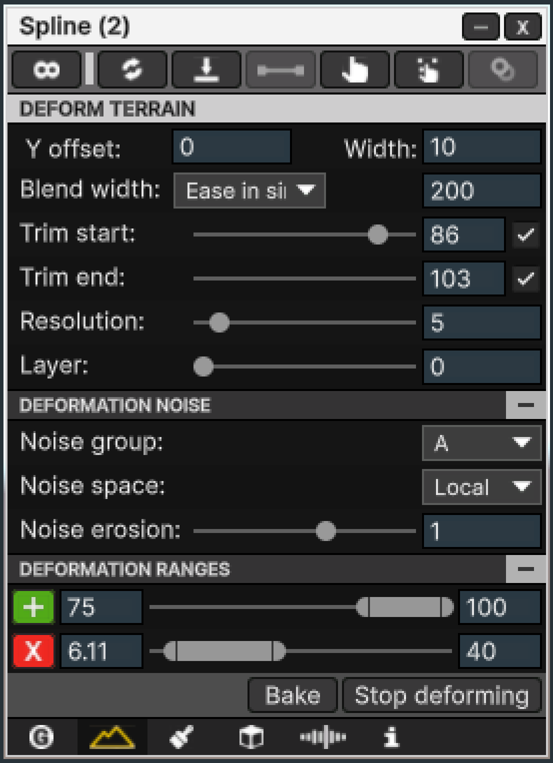
Terrain deformation and object instantiation has an indestructible work flow. Meaning you can always revert back to how it was before.

You can easily create whole worlds by creating mountains, rivers and path in this indestructible workflow. You do that by combining different noise layers and applying that to your terrain deformation, terrain painting and object instantiation.

Menus

When you install the Terrain tools addon you will get access to 3 sub menus for the spline.

Terrain deformation



You start deforming the terrain by pressing the "Start deforming" button inside the spline menu, now the terrain will update when you do any changes to the spline. To stop deform, press the same button again. You can also bake the terrain deformation created by a spline by pressing the "Bake" button. This will stop any further deformation updates from that spline and apply its deformation to the terrain's base layer.

If you deform a mountain with multiple paths on it, you should increase the layer value of the paths so Spline Architect knows which spline is on top of the others. Otherwise, the paths will disappear when moving the mountain or changing its values.

> ❗ Deforming the terrain at the bottom can cause unwanted deformation buildup. You can fix this by right-clicking on the Terrain component in the Inspector window and press "Raise Terrain by 10%".

Deform terrain

- **Y offset** | Extra height added between the spline and terrain (can be negative).
- **Width** | Width for the terrain deformation.
- **Blend width** | Determines the horizontal extent of terrain deformation, defining the transition area where the terrain shifts from its original to its deformed state. You can also change its curvature by setting different easings.
- **Trim start** | Trims the size of the start of the terrain deformation.
- **Trim end** | Trims the size of the end of the terrain deformation.
- **Resolution** | Increase the quality of terrain deformations. Try increasing this value if you get unwanted behaviours when deforming terrain.
- **Layer** | Sets the layer of the spline. Lower layers are deformed before splines with higher layers, so higher layers will always appear on top of lower-layered splines.

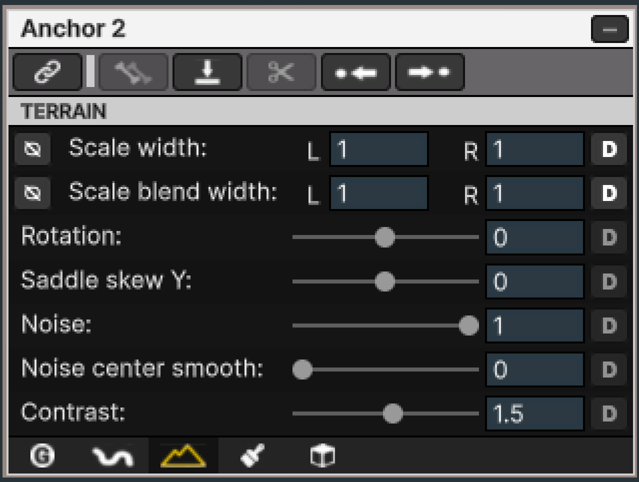
Deformation noise

- **Noise group** | The noise group you want to be applied to the terrain deformation.
- **Noise space** | What space the noise should originate from.
- **Noise erosion** | Warps the noise in an erosion-like way, either to the right (positive values) or to the left (negative values). This is not a true erosion simulation, but it can produce visually similar results with virtually none of the performance cost of a real calculation.

Deformation ranges

- **Range** | The range of the spline that should deform the terrain. You can have up to 16 range values on the same spline.

Terrain deformation effects

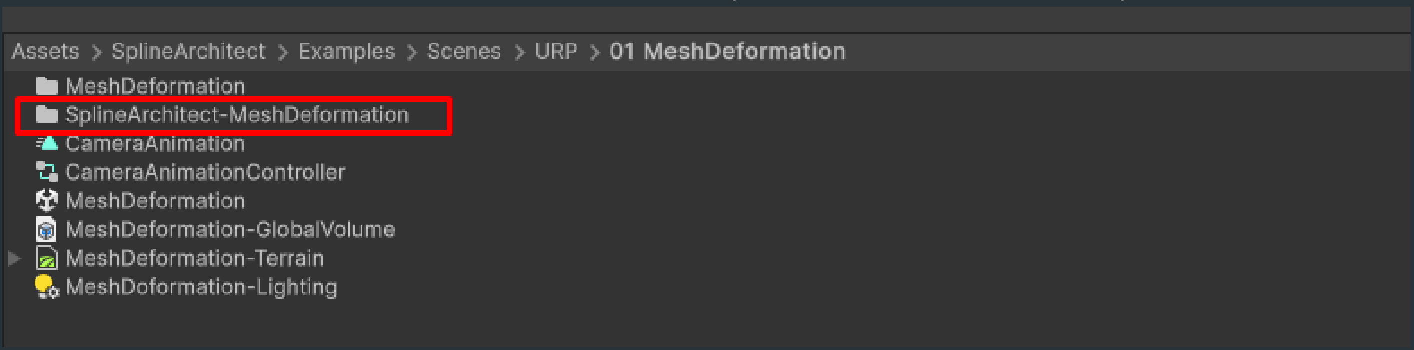


Selecting control points give you access to the menu above, where you can apply different behaviours for each control point on how you want to deform the terrain.

> If you want to use values outside the slider range, manually enter the desired value in the input field.

Terrain data

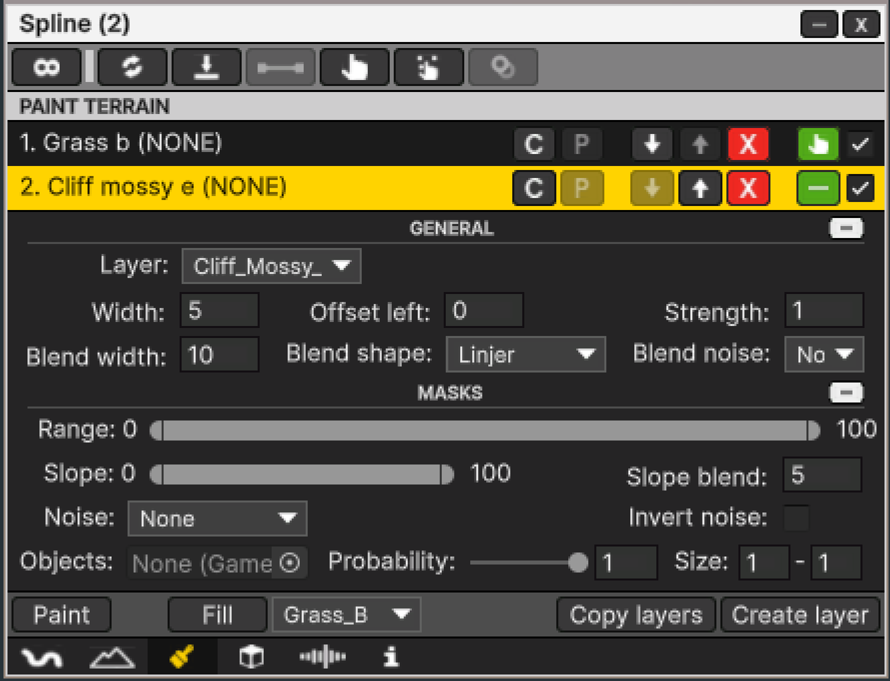
Spline Architect generates binary files during terrain deformation. These files store terrain data, allowing you to work with a non-destructive workflow. If you delete this data, you may experience unwanted terrain build-up. These binary files are located inside the 'SplineArchitect-YourSceneName' folder. This folder will automatically be created when necessary.



> ! If you rename your scene file you need to rename the SplineArchitect data folder to 'SplineArchitect-YourNewSceneName'.

Terrain painting

You can have up to 16 paint layers for each spline. The order of each layer matters where the layer 1 will be painted first, 2 will be painted second and so on. You can order them by pressing any of the arrow keys on the specific layer. You need to have one or more layer palettes assigned to the terrain for creating layers.



General

- **Layer** | The terrain layer pallet you want to paint with.
- **Width** | The painted width that originates from the spline.
- **Offset left** | Offset the painting to the left or right by adding a negative or positive value (the offset is in Unity meters).
- **Strength** | The strength of the layer palette related to other palettes. The value can be above 1 and some times this is preferable.
- **Blend width** | The painted blend width that originates from the spline.
- **Blend shape** | Applies an easing function that modifies the shape of the blend width.
- **Blend noise** | Applies a noise pattern to the blend width, causing it to vary based on the noise value.

Masks

- **Range** | The range of the spline that should be painted.
- **Slope** | Will only paint within this slope value.
- **Slope blend** | How strong the blending of the slope painting will be.
- **Noise** | Applies a noise mask.
- **Invert noise** | Inverts the noise mask.
- **Objects** | Assign a GameObject to this field. All of its child objects will act as a mask, restricting painting to their positions.
- **Probability** | The likelihood that a mask object's position will be used.
- **Size** | Random size for all object masks. The final size also scales with the original object's transform.

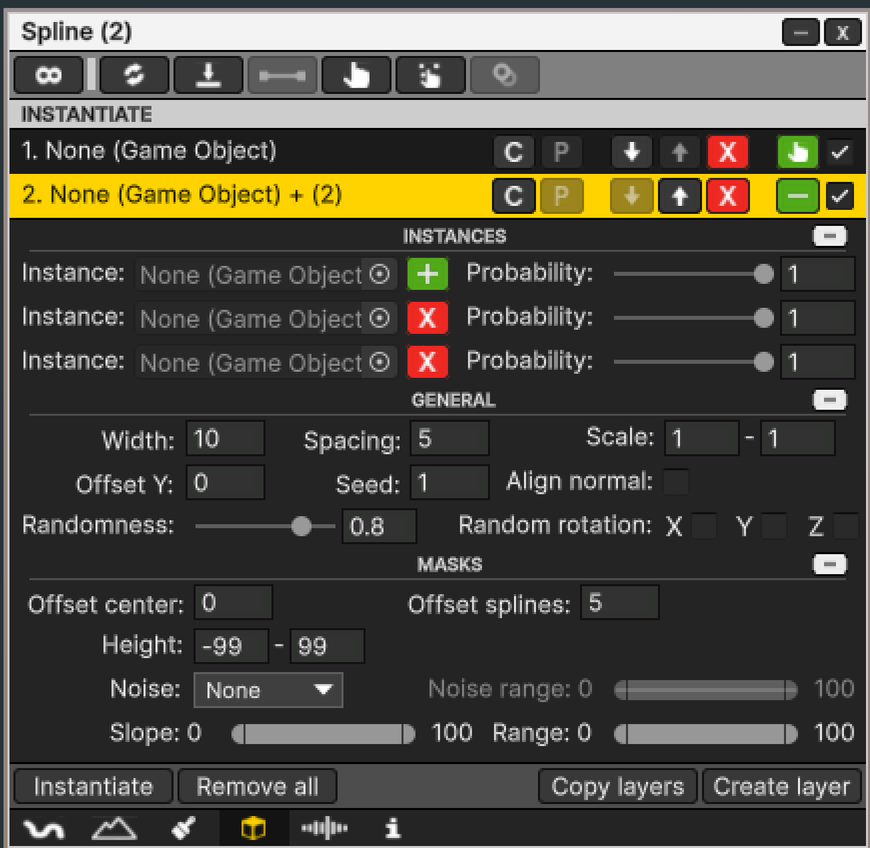
Terrain painting effects

Selecting control points give you access to the menu above, where you can apply different behaviours for each control point on how you want to paint the terrain.

> If you want to use values outside the slider range, manually enter the desired value in the input field.

Object instantiation

You can have up to 16 instantiate layers for each spline. The order of each layer does not matter. But you can change order if you want.



Instances

- **Instance** | Drag a prefab or any GameObject in the hierarchy you want to instantiate. You can create up to 16 instance fields in each layer.

- **Probability** | The probability setting for each instance field. 1 is the highest, if you set it to 0 it will never be instantiated.

General

- **Width** | The width of the instantiate are. Originates from the spline.
- **Spacing** | The spacing between each instance.
- **Scale** | Random scale for instances.
- **Offset Y** | Offsets all instances in the Y axel, can be negative.
- **Seed** | A number that controls all randomness. Using the same seed will always generate the same result. Changing the seed produces a different random arrangement.
- **Align normals** | Will align the instance transform with the terrain or whatever the GameObjecet should be instantiated on.
- **Randomness** | The randomness factor.
- **Random rotation** | Random rotation for specific axels.

Masks

- **Offset from center** | Skip instancing in a set distance from the center of the spline. Can be useful if you want to deform a path in the terrain, but only want to instantiate trees (for example) on the side of the path.
- **Offset from splines** | Of close instances can be other other splines (not the one your instancing from) in the scene.
- **Height** | The height range instances can spawn within. Originates form the spline.
- **Noise** | The noise group you want to use for all instances.
- **Noise range** | The range of the noise value where instances can spawn.
- **Slope** | What slope value instances can be spawned on. Having for example 0 to 20, will only spawn instances on nearly flat to flat ground.
- **Range** | The range of the spline, instances can be spawned from.

Object instantiation effects

Selecting control points give you access to the menu above, where you can apply different behaviours for each control point on how you want to instantiate objects along the spline.

> If you want to use values outside the slider range, manually enter the desired value in the input field.

Object Cloning

The **Object Cloning** menu can be accessed from the menu of a selected SplineObject.
Here you can either clone the selected SplineObject or clone all of its children. The **Clone** button is only available when the GameObject has no children. If it has children, only **Clone Children** will be available.

To clone a sequence of GameObjects, start by creating an **empty GameObject** and parent it to a spline. Then parent any followers or deformations to this empty GameObject. Next, open the **Object Cloning** submenu for the empty GameObject and press **Clone Children**.

Clones cannot be moved using the position tool or the position fields in the SplineObject menu. This is because they are always aligned with the offset value. However, you can **disconnect** individual clones or the entire clone section. Disconnect a clone by pressing **Disconnect** in the clones menu, or disconnect the entire section by pressing **Disconnect** on the clone parent.

You can clone any deformation or follower either to the **end of the spline** or by specifying a **fixed number of clones**. You can also clone a sequence of deformations and/or followers. When no fixed amount is specified, cloning continues to the end of the spline and the number of clones updates automatically.

If **Snap End** is enabled, the **last clone** will snap to the end of the spline and completely fill it.